

How we build it

AI Video Editor

A web video editor with a real timeline, where the AI tools sit in the toolbar next to the ordinary ones. This page shows **how the system is put together** and how it pays for itself.

EDITOR CONFIRMED

PHASED RELEASE

WEB FIRST

PRICING AGREED

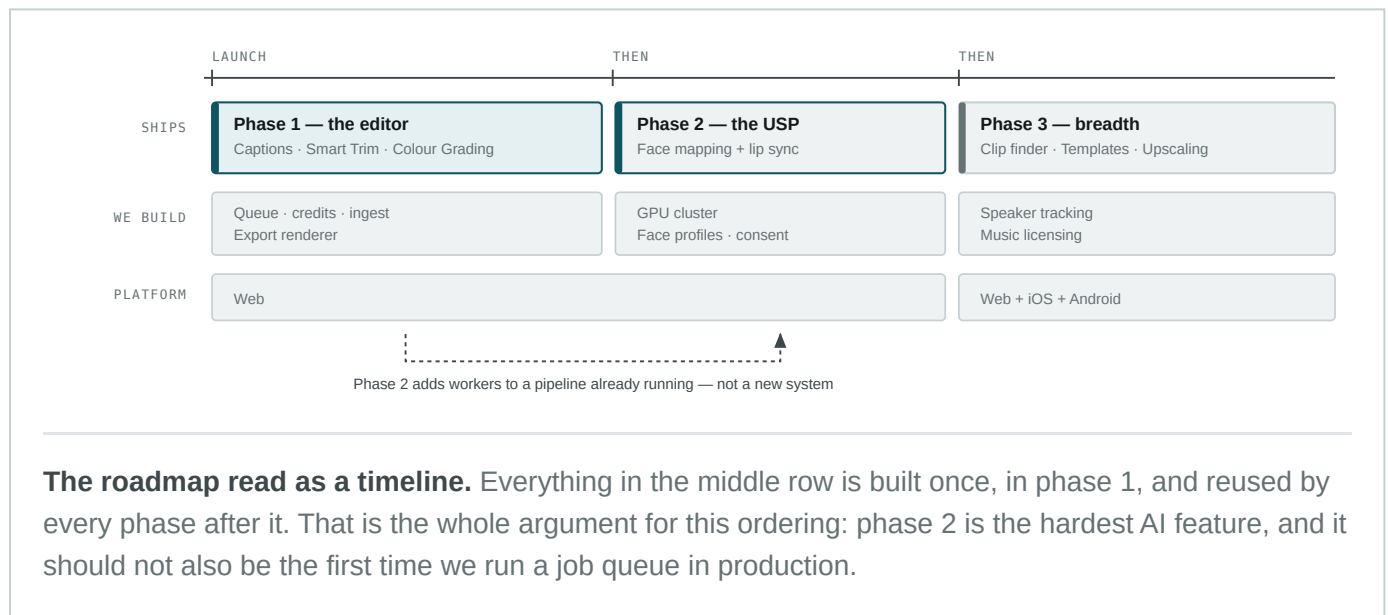
NOTHING BLOCKING

13 AUGUST 2026

THE PLAN

Three phases, each building on the last

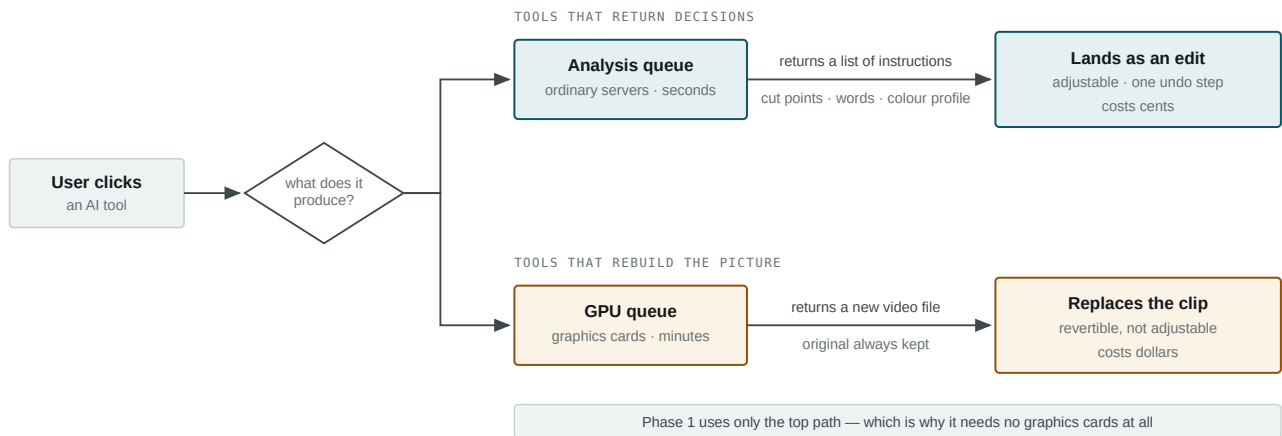
Phase 1 ships a working editor with the three AI tools that cost least and help most. The differentiator — face mapping — arrives in phase 2, onto machinery that is already running in production rather than being built alongside it.



THE KEY IDEA

AI tools come in two kinds, and it decides almost everything

Some tools can tell us **what to do** — where to cut, what was said, which colour profile fits. Others have to rebuild the picture itself. Those two things cost different amounts, take different hardware, and leave the user with different amounts of control. Sorting every tool into the right one is the single most consequential decision in the design.



Same button to the user, completely different machine underneath. Captions, smart trimming and colour grading all take the top path: they analyse and report back, cheaply, and the user can still change anything afterwards. Face mapping has no choice but the bottom path — there is no way to describe a replaced face as an instruction.

TOP PATH — PHASE 1

Cheap, fast, still editable

The server sends back a list. The editor turns it into ordinary cuts and captions that the user can drag, retype or undo.

TOOLS Captions, Smart Trim, Colour Grading

TAKES Seconds

NEEDS Ordinary servers

BOTTOM PATH — PHASE 2

Slow, expensive, worth it

The server rebuilds the footage frame by frame and hands back a new file. The original is kept, so it can always be undone.

TOOLS Face mapping, lip sync, noise removal

TAKES Minutes

NEEDS Graphics cards

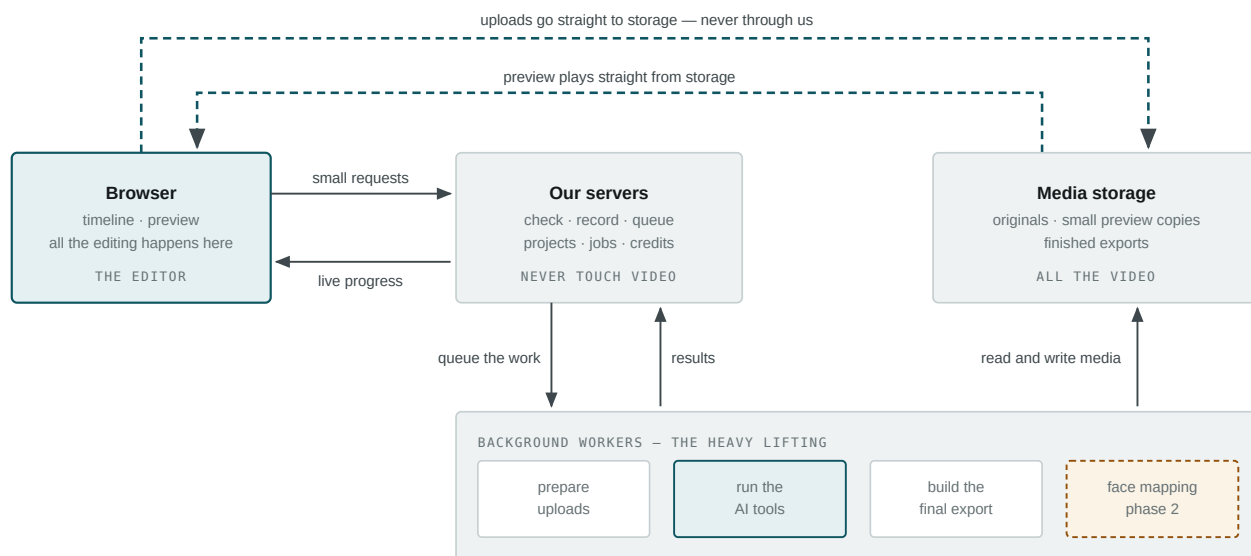
WHY THIS SAVES REAL MONEY

Graphics cards are rented by the hour whether they are working or not. By putting every phase 1 tool on the top path, we launch without renting any — and the cluster arrives in phase 2 alongside the one feature that genuinely needs it.

THE SYSTEM

What talks to what

The part worth noticing: video files never pass through our application. The browser uploads straight to storage and plays straight from the delivery network. Only small control messages — "make me a project", "run captions on this clip" — touch our servers.

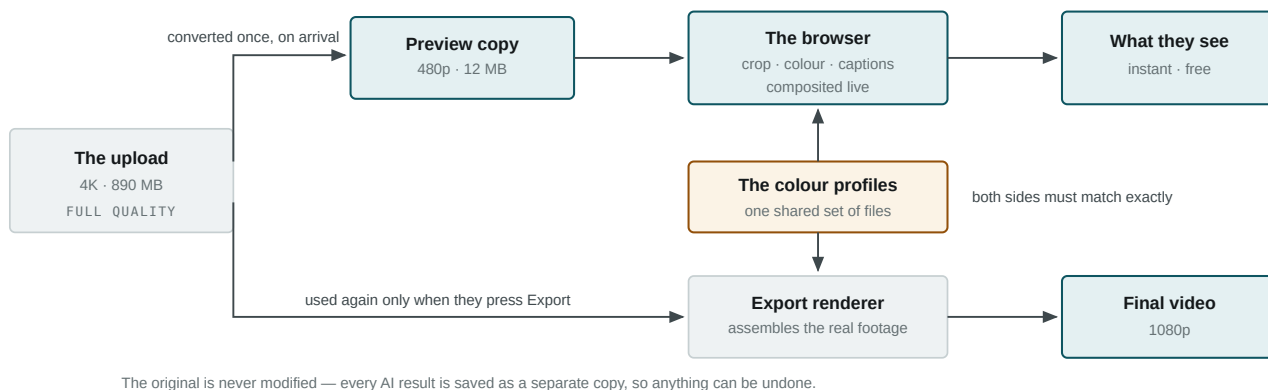


Dashed teal lines are video; solid lines are messages. Keeping video off our servers is what stops bandwidth costs scaling with usage — and it is why uploading a 900 MB file is as reliable as any other file transfer, rather than depending on our application staying up for ten minutes.

WHY IT FEELS INSTANT

Two copies of every video, doing two different jobs

When someone uploads footage, we immediately make a small, light copy of it. That copy is what they scrub, cut and preview — which is why editing responds instantly and costs us nothing. The full-quality original is touched exactly once more: at export.

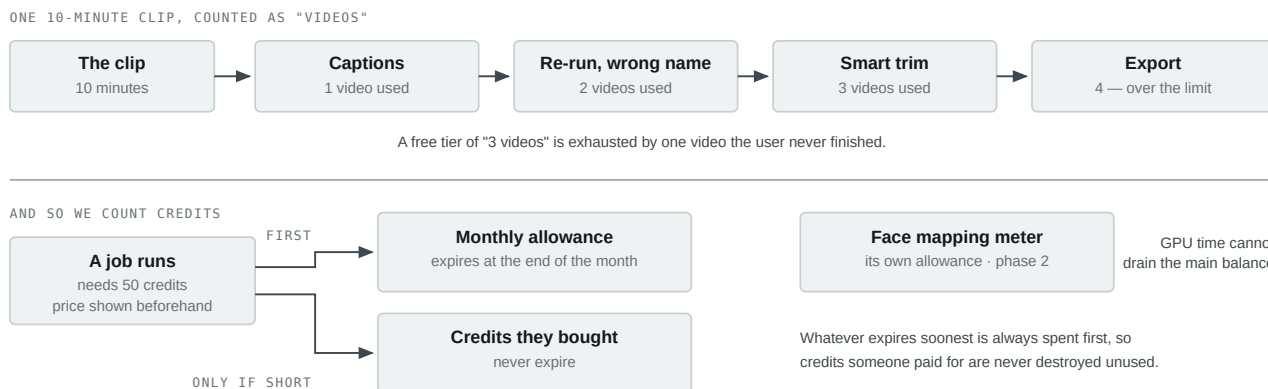


The amber box is the part that has to be watched. The browser and the export renderer are two different pieces of software applying the same colour treatment. If they ever drift apart, someone edits a video that looks one way and downloads one that looks another — so they share the same profile files and are tested against each other.

HOW IT PAYS FOR ITSELF

We show videos,
we count credits

Tiers are advertised as "≈30 videos a month" because that is what a creator understands. Underneath, everything is metered in credits — because a literal video counter breaks the first time someone re-runs captions on a clip they have not finished.



The top row is the failure we designed around; the bottom row is the fix. Credits meter the actual work, so re-running captions costs a few credits rather than a whole video, and each tier carries 25–40% more credits than its headline figure needs to absorb exactly that. The "≈30 videos" stays on the pricing page, where it belongs.

WHERE THE MARGIN IS COMFORTABLE

On phase 1 tools — captions, trimming, grading, export — a ten-minute video costs us roughly **\$0.10–0.15** to process. Pro at \$19.99 for thirty videos leaves plenty of room.

WHERE IT NEEDS WATCHING

Face mapping runs on graphics cards and is an order of magnitude more expensive. That is precisely why it has its own metered allowance instead of sharing the main balance.

THESE ARE ESTIMATES UNTIL WE MEASURE THEM

The cost figures above are calculated, not observed. Cost-per-job is instrumented from the first deployment, so the tiers get re-priced on evidence rather than arithmetic — and because plans live in a database table, changing a price is a data change, not a release.

WHERE WE STAND

**Nothing is blocking.
Building can start.**

Every decision the architecture depends on has an answer. What remains has a known default and is needed later, not now.

1 Open the Stripe and Razorpay accounts

Both providers go live at launch, and both need business verification before they will process a real payment. Razorpay additionally needs Indian entity documentation.

Not blocking today — billing is built and tested against test keys. But the lead time is external and nobody on the team controls it, so the applications should start now rather than when the code is ready.

2 Smart trimming: tighten a recording, or cut it down to its best parts?

Removing silences and stumbles saves roughly 10–25% of a recording. The original spec promises ten minutes down to two, which means choosing what is worth keeping — a substantially harder tool.

Affects what we promise publicly, not what gets built first. Needed before the feature is described to customers.

3 Storage retention, and who owns tax

How long we keep uploads is the largest recurring cost in the system and the only one that grows on its own. Indian GST and EU VAT both need configuring — the providers offer tools, but choosing and setting them up is not a development task.

Both needed before launch, neither blocks building.

ONE THING TO KEEP IN VIEW

Phase 1 ships without face mapping, so it cannot be marketed as the face-swap product — its story is "the editor that does the boring work for you". That is the right build order, but if launch is being sold on the differentiator, it is a conversation worth having now rather than in three months.

THE FULL SET

Where the detail lives

This page is the overview. Everything below is written and in the repository, ready for the development team.

01

Product Vision

What the product does, every decision made and every one still open

02

Phase 1 Scope

The cold list of what ships now, and what explicitly does not

03

Backend Architecture

Data model, job pipeline, media, rendering, billing, infrastructure

04

Frontend Architecture

Timeline state, the playback engine, undo, tool integration

05

API Contract

The interface both teams build against, in full

—

Diagrams

System overview, data model, job and subscription lifecycles

AI Video Editor · Technical Architecture Overview · 12 August 2026 · MMaxouB

Built on the project lead's decisions of 12 August 2026. Phase 1 scope is firm; the AI tool selection is a proposal awaiting decision A.

Full documentation set in `docs/`.